



Interim Report

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Table Of Contents

1. Introduction	1
1.1 Problem scenario	1
1.2 The problem as a solution	2
1.3 Aims and Objectives	2
1.4 Resources required	2
2. Background	5
2.1 Review of articles and journal	5
2.2 Related projects	10
2.3 Comparison between different application	12
3. Work done	14
3.1 Flowchart Diagram	14
3.2 ERD	15
3.3 Use Case Diagram	16
3.4 Collaboration Diagram	17
3.5 Sequence Diagram	18
3.6 Backend	18
3.7 Wireframe	21
4. Methodology	21
4.1 Different Methodology	22
4.2 Work breakdown structure	24
4.3 Milestones	26
4.3.1 Revised Milestones	28
4.4 Project Gantt Chart	28
5. Further Work	30
6. References	32

Table of Figure

Figure 1	3
Figure 2	3
Figure 3	4
Figure 4	4
Figure 5 E-Commerce Web Application by Using MERN Technology	6
Figure 6 An Overview of Electronic Commerce e-Commerce.....	7
Figure 7 The Role of e-Commerce	9
Figure 8 Barber supply	10
Figure 9 Barber Blades.....	11
Figure 10 Barber Depots	12
Figure 11 Flow Chart Diagram.....	14
Figure 12 ERD.....	15
Figure 13 User Case Diagram.....	16
Figure 14 Collaboration Diagram.....	17
Figure 15 Sign up	19
Figure 16 Login.....	20
Figure 17 Wireframe	21
Figure 18 Sequence Diagram.....	21
Figure 19 Agile Method.....	22
Figure 20 Waterfall method	23
Figure 21 Increment method.....	24
Figure 22 Revised Gantt Chart.....	29

1. Introduction

In the current era of technology, there has been a significant increase in the need for online services, with the barbering sector also experiencing this trend. Our new platform, Yanki Barber is ready to completely change how customers interact with barber products. Yanki Barber is set to function as an all-inclusive virtual platform, effortlessly linking customers with a diverse selection of barbers. Users can explore a wide range of services, check out detailed portfolios, and read reviews to help them make informed choices that suit their individual preferences and requirements. It will include a built-in online shopping section with a carefully chosen variety of top-notch products items from reputable brands, making it convenient for customers to find all they require for their hair care and grooming needs. Yanki Barber seeks to improve customer convenience and satisfaction by simplifying the process of locating barbering services and offering a platform for buying high-quality hair care products, ultimately revolutionizing the grooming industry.

Current Scenario: The current barbering products market is divided, requiring customers to go to various stores to find what they need. This process may consume a lot of time and be inconvenient. There is an increasing need for a unified online platform that provides a wide variety of barbering products, as well as detailed product descriptions and reviews. Barbera meets this demand by offering a convenient online platform exclusively for barbering and hair care products. **Aims and Objectives**

1.1 Problem scenario

Customers encounter difficulties when it comes to finding a single platform that provides a wide variety of premium barbering products. Current e-commerce platforms are frequently too standard, which makes it challenging for customers to find specific products that meet their grooming requirements. This leads to a disjointed shopping experience and customer discontent.

1.2 The problem as a solution

Yanki Barbera plans to address these issues by developing a unique online marketplace specifically for barber supplies. The website will showcase a diverse range of hair care, grooming, and barbering tools from multiple well-known brands. Users can navigate categories, access thorough descriptions of products, check ratings and reviews, and conveniently compare items. The platform aims to offer a smooth and easy shopping experience, boosting customer satisfaction and loyalty.

1.3 Aims and Objectives

a) Aims:

To simplify the purchasing of barbering supplies, we plan to create a website with a diverse selection of products from various brands to accommodate our customers' varied preferences and requirements. The website will offer easy-to-use tools like comparing products, reading user reviews, and accessing detailed product information to improve the shopping process. Moreover, the platform will support a barbering-focused community by promoting user interaction with reviews and ratings.

b) Objectives:

- Create a website interface that is easy for users to navigate and search for products.
- Collaborate with a range of brands to offer a diverse range of barbering supplies.
- Incorporate secure payment gateways to guarantee secure transactions.
- Develop effective logistics strategies to guarantee punctual and dependable delivery of products.

1.4 Resources required

a) Visual studio code

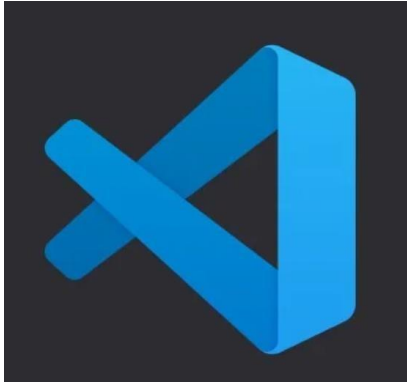


Figure 1

Visual Studio Code is a lightweight yet potent source code editor that is free and can be used on desktop or web platforms. It comes with diverse ecosystem extensions for various programming languages (such as C++, C#, Java, Python, PHP, and Go), runtimes (such as .NET and Unity), environments (such as Docker and Kubernetes), and cloud platforms (such as Amazon Web Services, Microsoft Azure, and Google Cloud Platform). (Murali, n.d.)

b) Figma

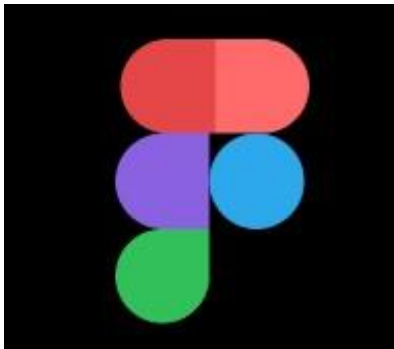


Figure 2

Figma is a design tool in the cloud that is used for web and app development and testing. Despite offering similar features, its unique selling point is its ability to facilitate real-time collaboration among team members, distinguishing it from competitors such as Sketch and Adobe Xd.

c) Nodes.js



Figure 3

Nodejs is an open source, cross platform runtime environment that allows to execute JavaScript code on server side. With Nodejs JavaScript can also be used on server side. It is built on the V8 JavaScript engine from Chrome, which compiles JavaScript into efficient machine code.

d) Reacts.js

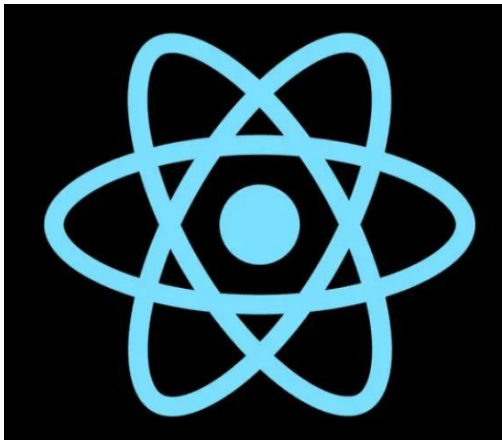


Figure 4

React.js is an open-source JavaScript library developed by Facebook for building user interfaces. In react web application is developed by creating reusable components. These reusable components are individual pieces of final interface which is assembled to form user interface.

2. Background

2.1 Review of articles and journal



[https://www.researchgate.net/publication/353481574 E-Commerce web Application by using MERN Technology](https://www.researchgate.net/publication/353481574_E-Commerce_web_Application_by_using_MERN_Technology)

The article "[E-Commerce Web Application by Using MERN Technology](https://www.researchgate.net/publication/353481574_E-Commerce_web_Application_by_using_MERN_Technology)" focuses on building an e-commerce platform using the MERN stack (MongoDB, Express.js, React.js, and Node.js). It highlights the design of a user-friendly web application with React.js for a dynamic frontend, Node.js and Express.js for a robust backend, and MongoDB for flexible data storage of products, users, and transactions.

The MERN stack's modularity enables seamless integration, rapid development, and scalability, making it ideal for startups. While React.js enhances interactivity, challenges like secure payment integration and user authentication are noted. The paper concludes that the MERN stack is a cost-effective and scalable choice for creating modern e-commerce platforms.

An Overview of Electronic Commerce (e-Commerce)

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Abstract: The world economy is witnessing a transition. All companies are transformed into information-based operations through online technologies. The pace of technical transition is so exponential that modern electronic commerce is now making significant shifts in the economic environment, impacting all areas of industry. The Web has expanded companies' scope. The vast quantity of business information made accessible by the global network that facilitates the gathering of information between firms, a corporation, its clients and the various divisions of a business is increasing exponentially. The information-based virtual value chains for any company cannot be overlooked operationally or strategically. This review article discussed the aspects of electronic commerce including its importance, facilitators, benefits, challenges and scope in the Indian market.

Keywords: Business, Consumer, Electronic commerce, Internet, Market strategy, Growth, Profit.

INTRODUCTION

Globalization as well as information technology (IT) change the method of the business doing by the organizations. In almost all companies that have invested largely in the IT infrastructure for the ultimate growth of their company, the IT system is implemented and integrated. The utilization of e-commerce (EC) as a way to execute transactions related to business is increasing concern. It has been a priority for many companies [1]. With EC, businesses will link "Just in Time production" and "Just in Time" to their trading partners, who boost their strategic abilities worldwide. EC's description is not widely agreed upon. From the communications point of view, EC may represent information, services as well products or online payments through telephone lines, computer networking, or other means from a communication point of view. EC implements technology for automating corporate transactions and workflows from a business process perspective. From a service viewpoint, EC is a way of lowering services costs while enhancing product efficiency and speeding up the delivery of services, which addresses industry, customers and management's desire. EC offers online shopping and distributing goods and information for the Internet and other online resources from an online point of view. Because of Internet and network technology's popularity and accelerated expansion, the electronic industry has become a significant field for contemporary enterprises. Large company operations are being carried out online today. People sell and purchase products and services online, and without internet infrastructure, certain purchases cannot be done [2].

This review article provides an overview of electronic commerce, mainly focused on its definition and why it is important for the modern market. It also discussed the different types of electronic commerce fields and their facilitators. Apart from the benefits offering by e-commerce, there are some disadvantages also that will be discussed in the subsequent sections. The last sections will discuss the trend and future of electronic commerce in India.

E-Commerce and Its Importance

E-commerce is referred to as electronic commerce. It means the electronic media and the internet for dealing with goods and services. E-Commerce entails a company accessing the internet as well as IT, such as the electronic data interchange (EDI). E-commerce concerns an internet vendor's website, trading goods or services to the user directly from the platform. The gateway uses a wireless purchase cart or a purchase basket to pay by credit card, debit card or Electronic fund transfer (EFT). A further description is as follows: Electronic communications and digital information processes in business transactions are used to create, modify and redefine value generation relations between, and between, organizations and individuals [3].

With the increasing spread of ICTs, specifically the Internet, the global corporate world pushes rapidly into e-commerce (Business-to-Business). As the Internet enables consumers to enter the global economy, they can compare prices across areas, find out how they vary by request, and become aware of substitution. The buyers obtain a distinct advantage. Thanks to market openness, consumers can conveniently compare e-commerce offerings from different websites. The rivals would immediately be one click away from the customer if the company is electronic. If consumers aren't comfortable with certain e-goods, content's pricing or services, they

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Figure 6 An Overview of Electronic Commerce e-Commerce

<https://www.researchgate.net/publication/351775073> An Overview of Electronic Commerce e-Commerce

This paper provides a comprehensive overview of e-commerce, focusing on its types, applications, benefits, and challenges. It highlights how e-commerce has transformed business operations by enabling digital transactions, improving customer access, and reducing operational costs. The article also discusses technological advancements such as payment gateways, logistics, and customer data analytics that have boosted e-commerce growth.

The Role of e-Commerce: A Systematic Literature Review

<https://doi.org/10.3991/ijim.v16i13.30611>

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Abstract—The implementation of digitalization in industrial sector surely gives impact to several sectors, particularly in business. The development of information technology and rapid economic globalization have initiated the role of e-commerce in economic trading activities over the globe. It takes numerous of researches to initiate online new business. Therefore, it is important to reveal the existence of gap among previous researches. Systematic literature review method is applied to analyse the role of e-commerce in trading activity as well to provide the improvement for future researches. As many as 28 e-commerce related literatures are analysed comprehensively and systematically based on protocol review. The result of research confirms the opportunity for future research on e-commerce system that enables the integration on business processes. The supporting variables of e-commerce include business branding, social and economic development, efficient system e-commerce platform, framework application.

Keywords—e-commerce, platform, business process, systematic literature review

1 Introduction

The evolving era leads to versatile activities. From the way of thinking to the methods in keeping up the activities that are supported by digitalization. The implementation of digitalization provides impacts on several sectors, especially in business sector. Both electronic information technology and economic globalization have been developed rapidly that put e-commerce's role to predominate the economic activities in many countries [1]. One of the activities of business process is web-based purchasing, by buying the goods are posted by the company, which sell and send the products through the internet [2].

To align the recent technology era that supported by the internet, several innovations are merged to form a method, called as online shopping. It utilizes a platform that specifically designed to support business process. Several things that should be considered in e-commerce, among others is the requirements of development by adopting the effective risk management technique in maintaining certain aspects, such as artificial

<https://www.researchgate.net/publication/361911355> The Role of e-Commerce A Systematic Literature Review

This systematic literature review explores the evolving role of e-commerce in various industries, emphasizing its impact on consumer behavior and business strategies. The study identifies key drivers for e-commerce adoption, such as convenience, trust, and technological infrastructure. It also discusses challenges like cybersecurity risks and highlights future opportunities, including AI and machine learning integration to enhance personalization.

2.2 Related projects

a. Barber Supply

<https://www.barbersupply.com/>



Figure 8 Barber supply

Barber Supply is a U.S.-based e-commerce platform offering a comprehensive collection of barbering products, including trimmers, shaving kits, and hair care products. The platform targets both professionals and individuals, featuring well-known brands at competitive prices. It provides features like free shipping for orders over a certain amount and focuses on convenience for its customers.

b. Barber Blades

<https://barberblades.co.uk/>



Figure 9 Barber Blades

Barber Blades is a UK-based e-commerce platform that caters to barbers and hairstylists. It offers an extensive selection of barber tools, furniture, and consumables, such as clippers, scissors, capes, and grooming products. The website emphasizes affordability and often includes discounted items and bundles. It also provides educational resources like tutorials for using their products.

c. Barber Depots

<https://www.barberdepots.com/>



Figure 10 Barber Depots

Barber Depots is an online store specializing in a wide range of barber equipment and supplies. Their catalog includes clippers, trimmers, razors, and styling products. The platform focuses on professional-grade tools for barbershops and individual stylists, offering products from well-known brands. The website features basic e-commerce functionality, including product browsing, categories, and secure checkout options.

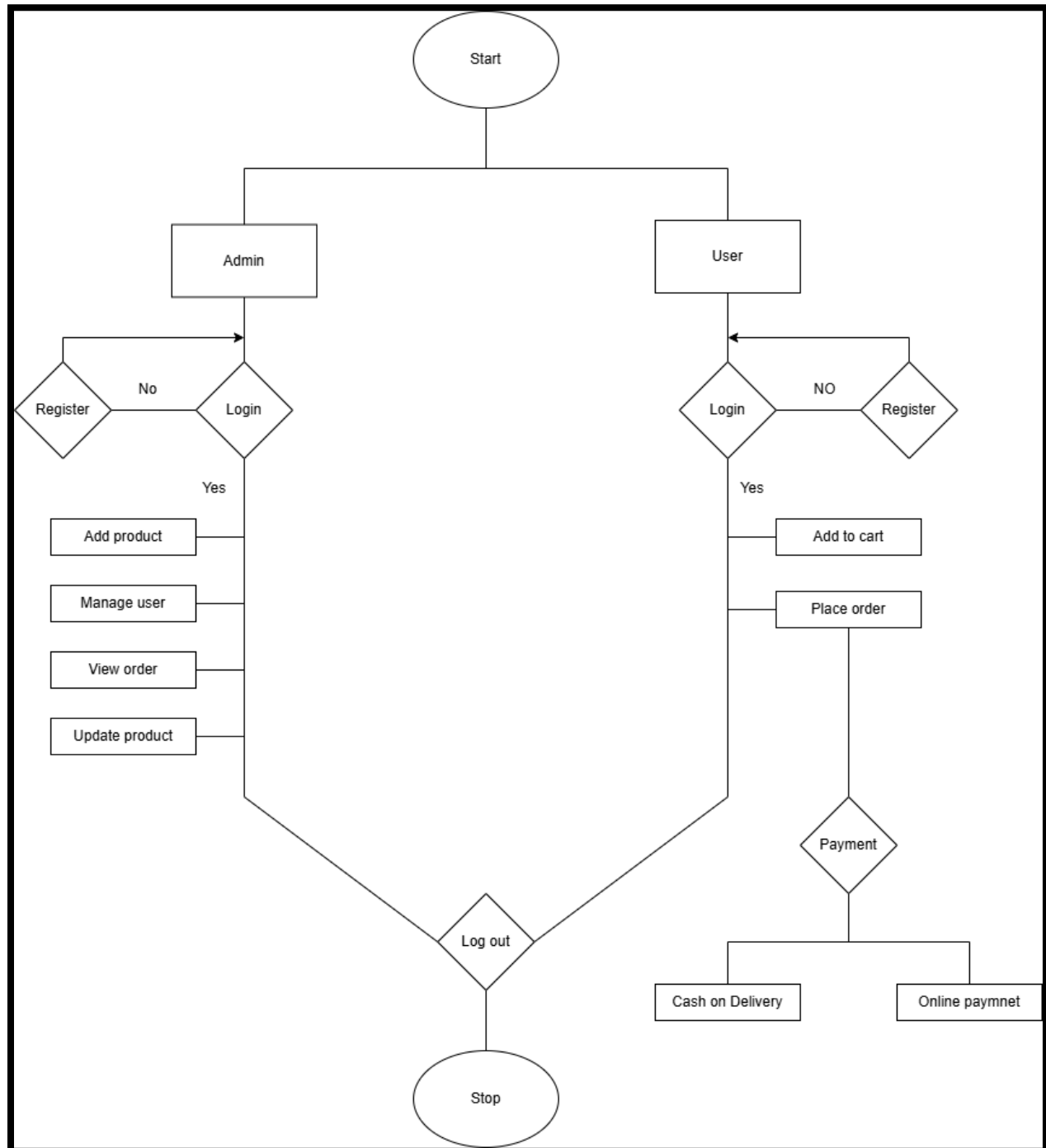
2.3 Comparison between different application

Feature	Barber Depots	Barber Blades	Barber Supply	My Project
Single vendor	False	False	False	True
Exclusive focus on barber equipment	False	False	False	True
Product Customization Options	False	False	False	True

Responsive Mobile Design	True	True	True	True
User Reviews and Ratings Feature	True	True	True	True
Secure Payment Gateway	True	True	True	True
Multi-Language Support	False	False	False	Optional
Product recommendation	True	True	True	True

3. Work done

3.1 Flowchart Diagram



S

Figure 11 Flow Chart Diagram

3.2 ERD

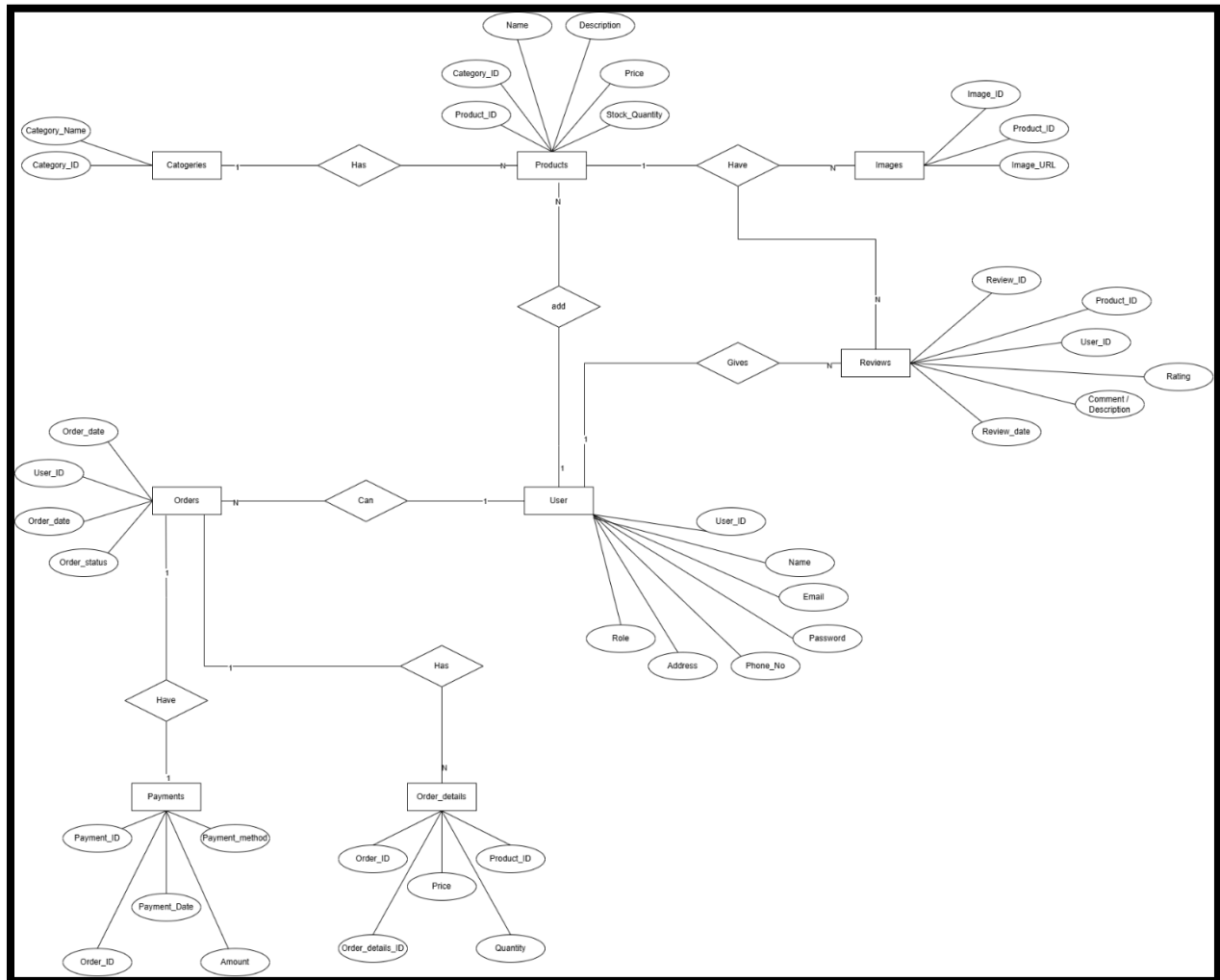


Figure 12 ERD

3.3 Use Case Diagram

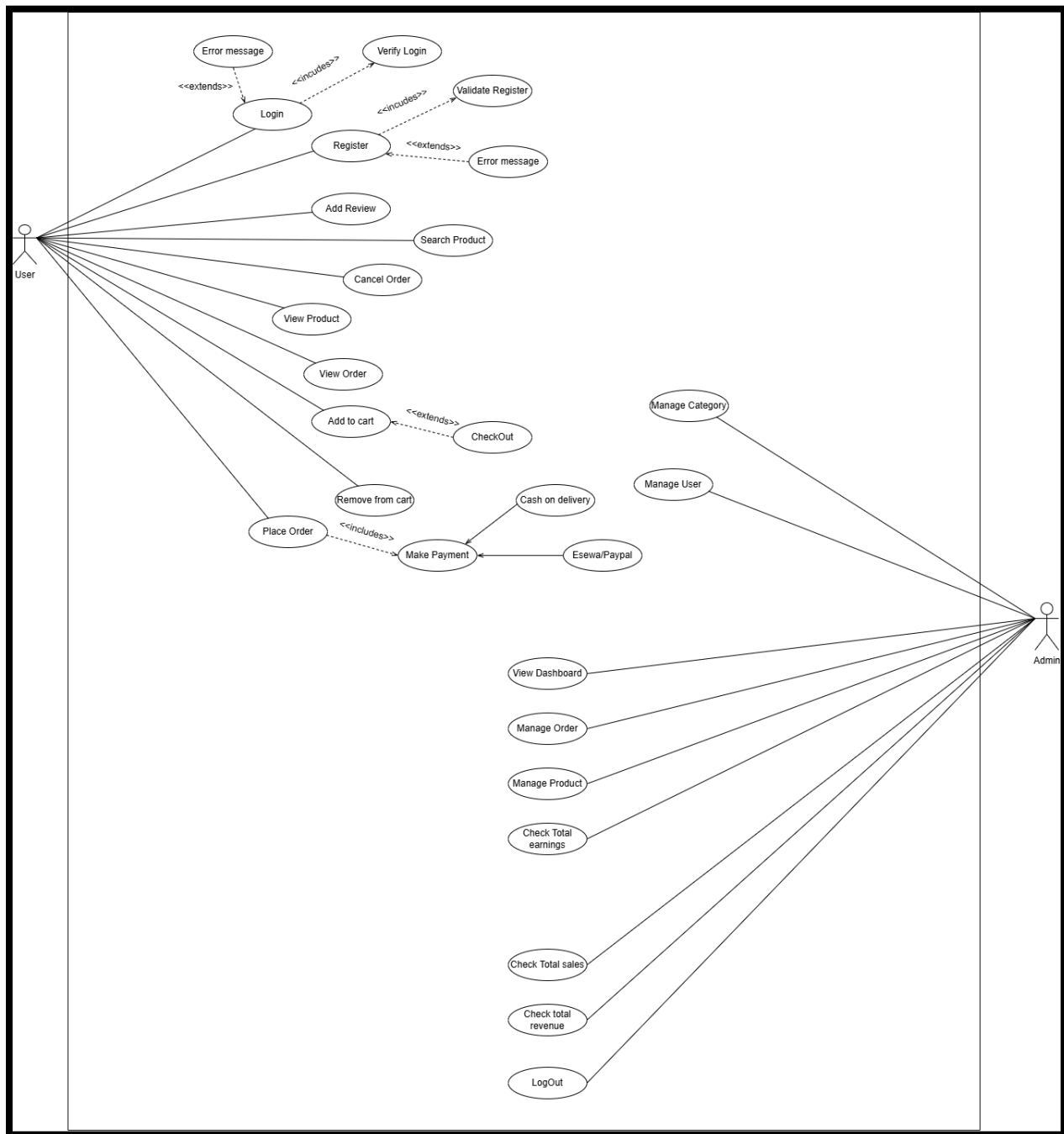


Figure 13 User Case Diagram

3.4 Collaboration Diagram

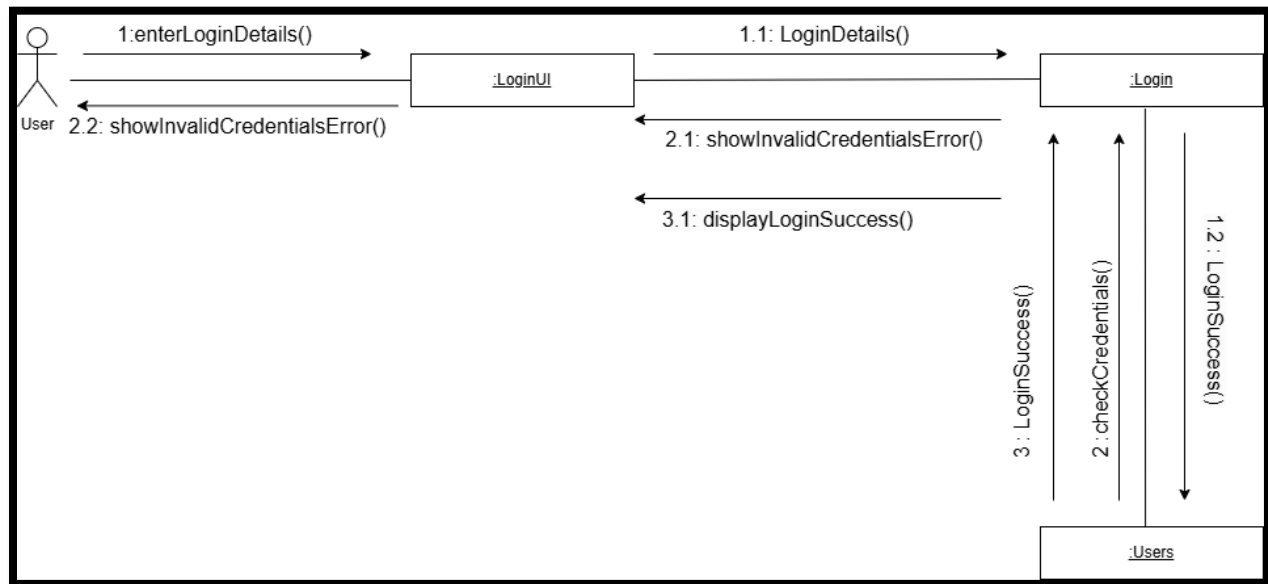
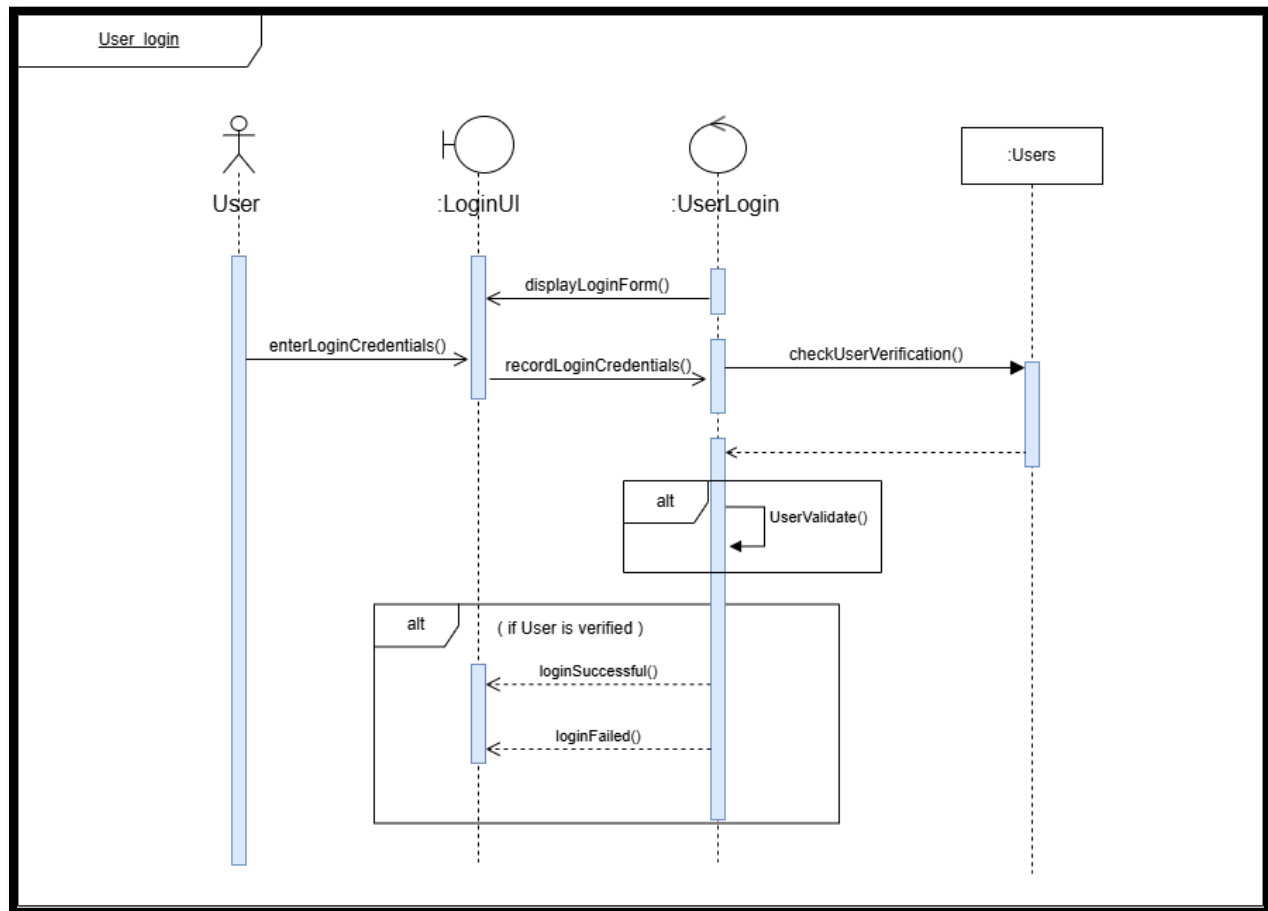


Figure 14 Collaboration Diagram

3.5 Sequence Diagram



3.6 Backend

Signup

```

8
9 const signSchema = Joi.object({
10   name: Joi.string().alphanum().min(3).max(30).required(),
11   password: Joi.string().alphanum().min(8).max(30).required(),
12   email: Joi.string().email().required(),
13 });
14
15 const signup = async (req, res, next) => {
16   try {
17     const { error } = signSchema.validate(req.body, {
18       abortEarly: false,
19       stripUnknown: true,
20     });
21     if (error) {
22       let errors = error.details.map((el) => {
23         return {
24           msg: el.message,
25           params: el.context.key,
26         };
27       });
28       return res.send(errors);
29     }
30
31     let hashedPassword = await bcrypt.hash(req.body.password, 10);
32     console.log(hashedPassword);
33
34     let user = await User.create({ ...req.body, password: hashedPassword });
35     let userObj = user.toObject()
36     delete userObj.password
37     console.log(userObj);
38     res.send(userObj);
39   } catch (error) {
40     console.log(error);
41     next(error);
42   }
43 }
44
45 module.exports = {
46   login,
47   signup
48 }

```

Figure 15 Sign up

Login

```
6
7 const loginSchema = Joi.object({
8   password: Joi.string().alphanum().min(8).max(30).required(),
9   email: Joi.string().email().required(),
10 });
11
12 const login = async (req, res, next) => {
13   const { error } = loginSchema.validate(req.body, {
14     abortEarly: false,
15     stripUnknown: true,
16   });
17   if (error) {
18     let errors = error.details.map((el) => {
19       return {
20         msg: el.message,
21         params: el.context.key,
22       };
23     });
24     return res.send(errors);
25   }
26
27   let user = await User.findOne({ email: req.body.email }).select({ "+password" : 1 });
28   if (!user) {
29     return res.status(400).send({ msg: "User not found" });
30   }
31   console.log(user);
32
33   let matched = await bcrypt.compare(req.body.password, user.password);
34   if (matched) {
35     user = user.toObject()
36     delete user.password
37     var token = jwt.sign(user, "shhhhh");
38
39     res.send({
40       user,
41       token,
42     });
43   } else {
44     res.status(401).send({ msg: "Invalid credentials" });
45   }
46 };
47
48
```

Figure 16 Login

3.7 Wireframe

Sign up

Full Name

Email

Password

Sign Up

Already have an account?Login

Figure 17 Wireframe

Figure 18 Sequence Diagram

4. Methodology

A software development methodology is a systematic method that emphasizes arranging and strategizing the process of constructing information systems. These methods are crucial in ensuring better quality software projects, assisting developers in the activities of software planning, development, and testing.

4.1 Different Methodology

a) Agile methodology:

Agile methodology is a project management strategy that enables the successful and efficient completion of a project, with a focus on enhancing the project and promoting teamwork within the team. The Agile approach in software development focuses on team cooperation and quickly delivering a functional product to satisfy customer requirements and anticipations. (Gurnov, n.d.)

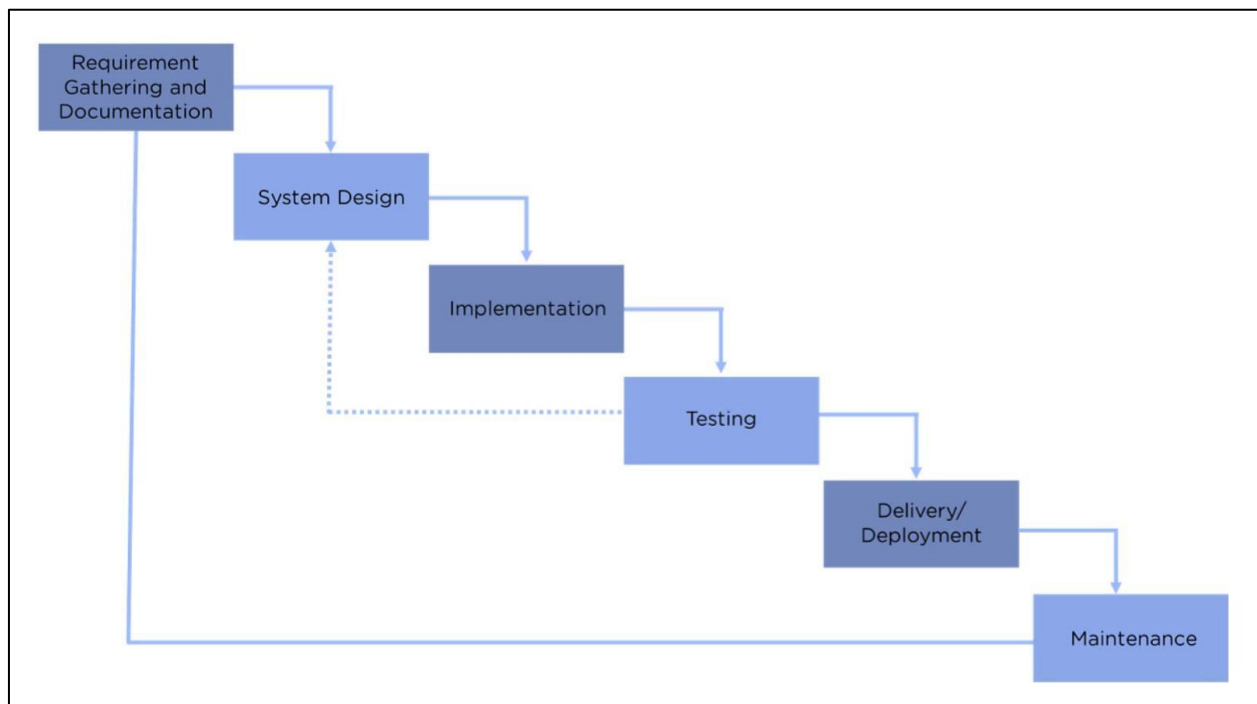


Figure 19 Agile Method

b) Waterfall methodology:

The waterfall methodology is a well-established project management procedure. Every stage of the process moves continuously downward through five steps, resembling a waterfall: requirements, design, implementation, verification, and maintenance. Unlike

other methods like Agile, Waterfall lacks flexibility. Before beginning the next phase, it is necessary to finish the current one. (Bottorff, 2024,)

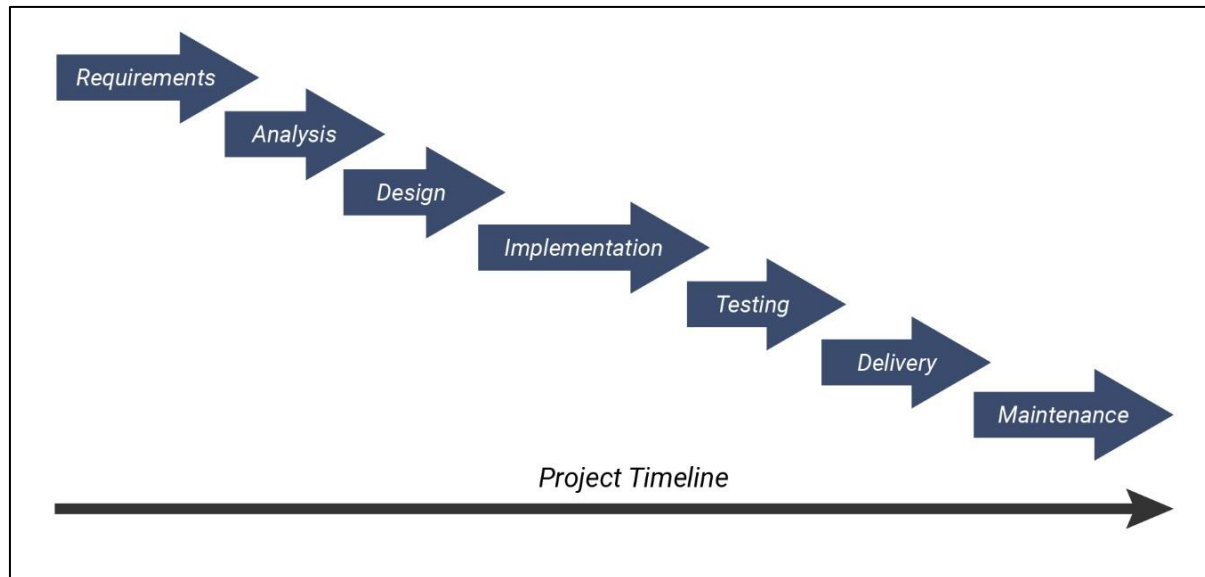


Figure 20 Waterfall method

c) Increment methodology:

In the Incremental model, software development processes are separated into various modules within the Software development life cycle (SDLC). This model could also be referred to as a Sequential version model. After splitting the modules, the incremental development process will commence in different phases. The waterfall model is used iteratively in every increment. The iterative approach also uses sequential progression in a prescribed order. (Claridge, 2020)

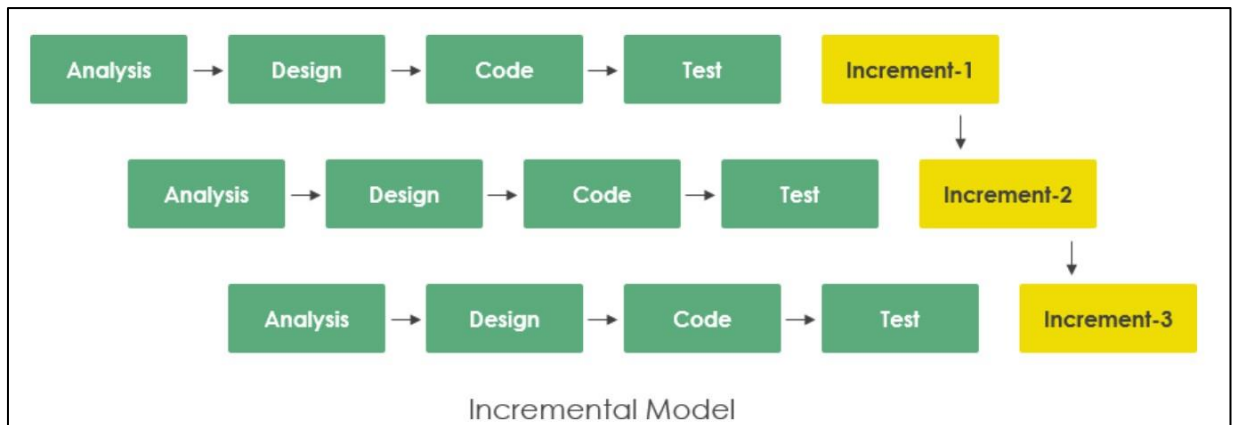


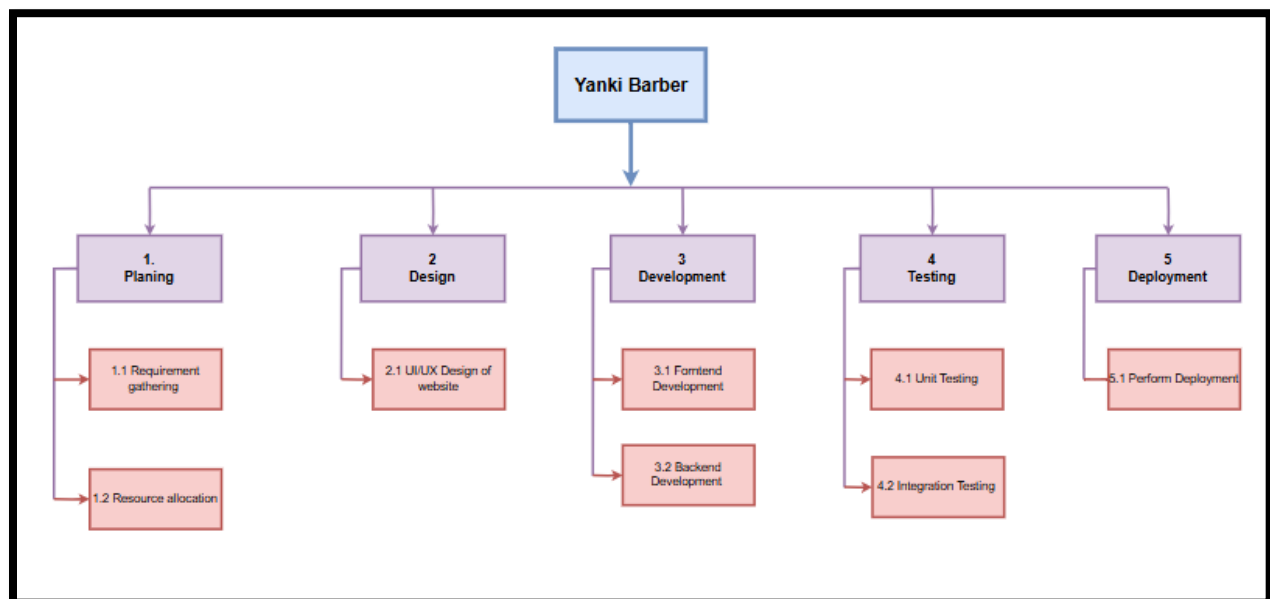
Figure 21 Increment method

Therefore, I am using the incremental model for my project as it is convenient and manageable for me. It is vital for this project to implement the incremental process as it allows for the systematic development of the system in manageable chunks, or increments. Each iteration represents a fully functional part of the system that is developed, tested, and delivered independently. By following this approach, the project will progress step by step, with each stage adding value and reducing complexity. By dividing tasks and improving concentration, the team delivers system components earlier than expected, allowing for early testing and feedback. Identifying and addressing potential issues in each increment boosts flexibility and enhances risk control. Ultimately, the incremental approach promotes enhanced project planning, easier project management, and more reliable and punctual project completion.

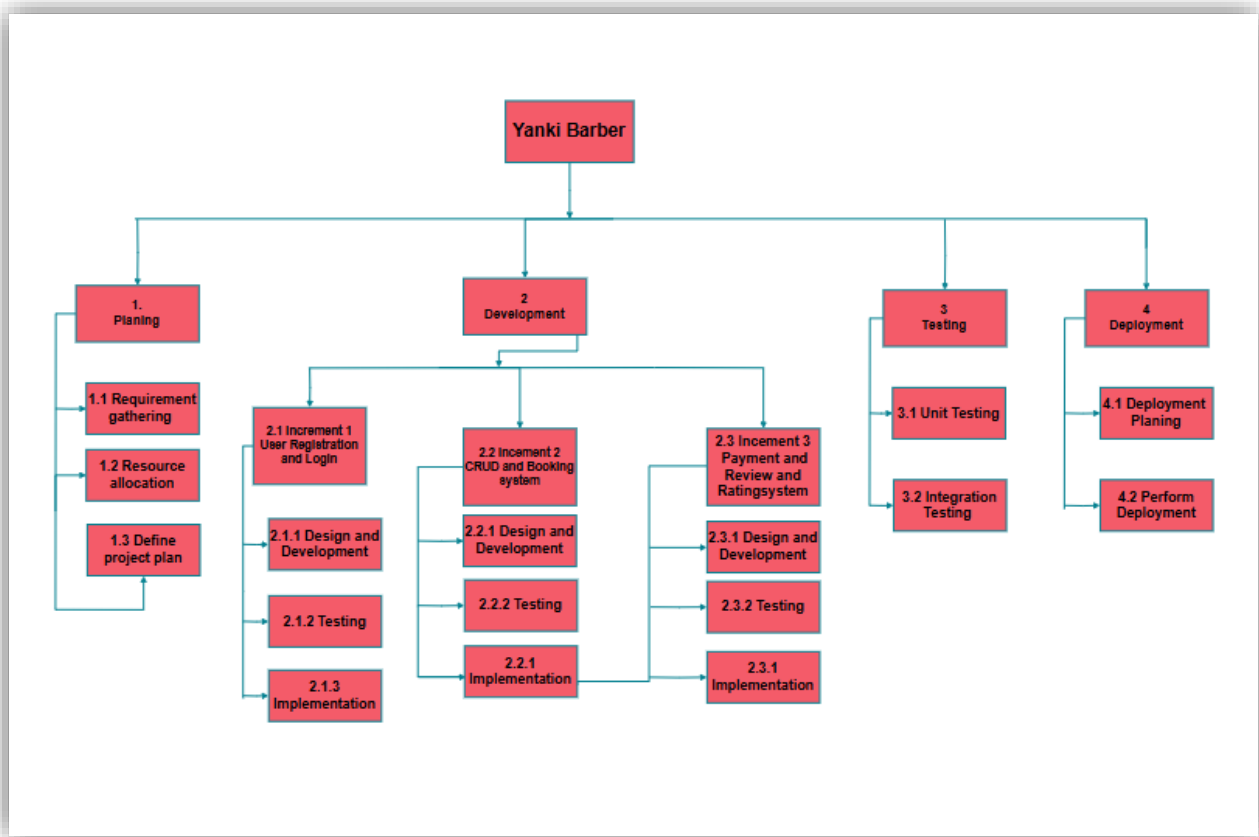
4.2 Work breakdown structure

Work breakdown structure (WBS) is a project management tool created to handle and finalize projects with numerous elements by dividing them into more manageable sections. (Organ, 2024) must be consulted for further information. WBS arranges the project in a hierarchical format. Assisting with project planning, scheduling, and monitoring tasks and deliverables. (Watts, 2024)

The "Yanki Barber" project's Work Breakdown Structure (WBS) is divided into five main phases: Planning, Design, Development, Testing, and Deployment. During the Planning stage, the project emphasizes collecting requirements and assigning essential resources. The Design phase includes developing the UI/UX for an interface that is easy for users to navigate. The Development phase is divided into Frontend and Backend responsibilities, with the frontend handling the design using tools such as React, and the backend dealing with server and database setup using Node.js. The Testing phase confirms the proper functionality of all components, both separately (Unit Testing) and together (Integration Testing). In the Deployment stage, the finished website is released and made available to users. This methodical strategy guarantees a systematic progression, resulting in the triumphant finalization of the Yanki Barber website.



Revised (WBS)



4.3 Milestones

A milestone is a significant event or point in a project's progress, schedule, or timeline that indicates the conclusion of an important stage or accomplishment. Milestones are employed in project management to monitor progress, verify that a project is meeting deadlines, and pinpoint key points that must be achieved before advancing to the next stage. Milestones help project managers to better predict the project timeline by emphasizing important dates and events.

are the key points for my project:

a) Requirement gathering will be completed by 26th September 2024.

b) Wireframe design in Figma will be completed by 12th October 2024.

- c) Login and registration system development will be completed by 16th November
- d) Frontend development will be completed by 1st November 2024.
- e) Analysis and testing of the user interface will be completed by 28th November 2024.
- f) Admin and host pages development will be completed by 14th December 2024.
- g) CRUD functionality will be implemented by 22nd December 2024.
- h) Search functionality will be developed by 5th January 2025.
- i) Analyzing, testing an implementation of multiple vendor login system by 26th January 2025.
- j) Order tracking system will be completed by 8th February 2025.
- k) Payment functionality integration will be completed by 17th February 2025
- l) System testing will be completed by 11th March 2025, with documentation prepared by 27th March 2025, and final submission on 30th March 2025.

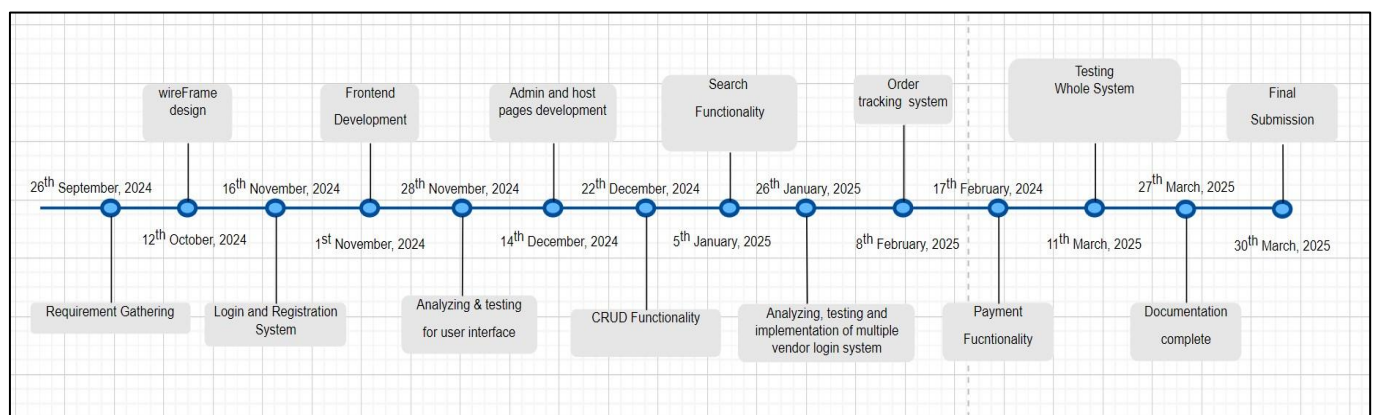


Figure 9 Milestones

4.3.1 Revised Milestones

Milestone	Description	Start Date	End Date
Planning	Define the scope, objective and timeline for the project.	2024-10-3	2024-10-23
Design	Finalize the architecture, UI/UX & complete the wireframe.	2024-10-24	2024-11-23
Development	Build the front-end & back-end components of the platform.	2024-11-24	2025-02-12
Testing	Conduct functional, performance & security testing; fix identified bugs.	2025-02-13	2025-03-25
Deployment & Launch	Deploy the project to production, launch the site, and begin user acquisition.	2025-03-26	2025-04-15
Post-Launch Support & Maintenance	Monitor performance, resolve issues & roll out updates as needed.	2025-04-16	2025-04-26

4.4 Project Gantt Chart

A Gantt chart serves as a visual representation of a project's timeline through bars on a chart. This visual aid, named after its creator Henry L. Gantt, is frequently utilized in project management for organizing, monitoring, and synchronizing tasks in a project. It presents a straightforward and summary of the planned tasks, their timing, and the responsible parties.

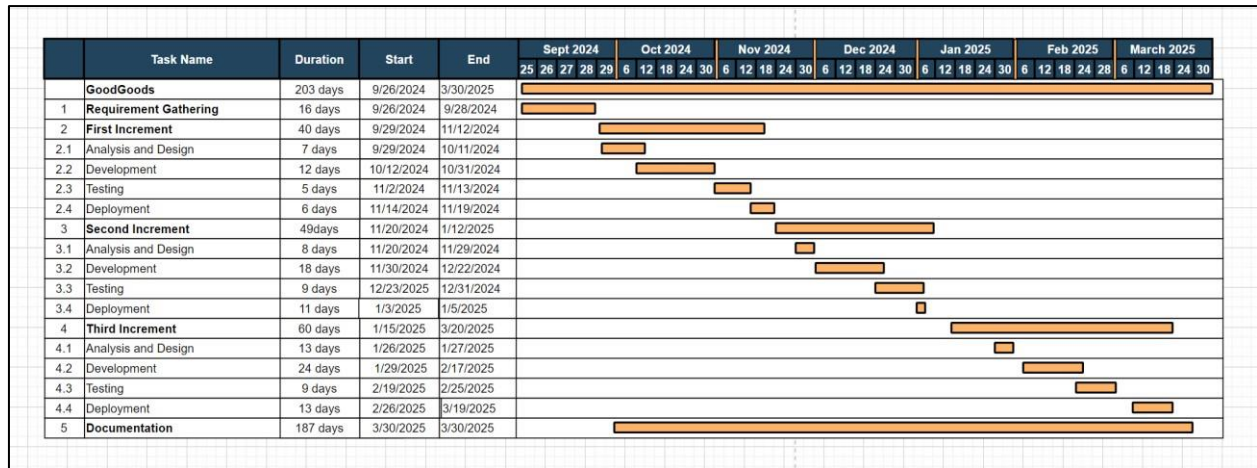


Figure 10 Project Gantt Chart

Revised Gantt Chart

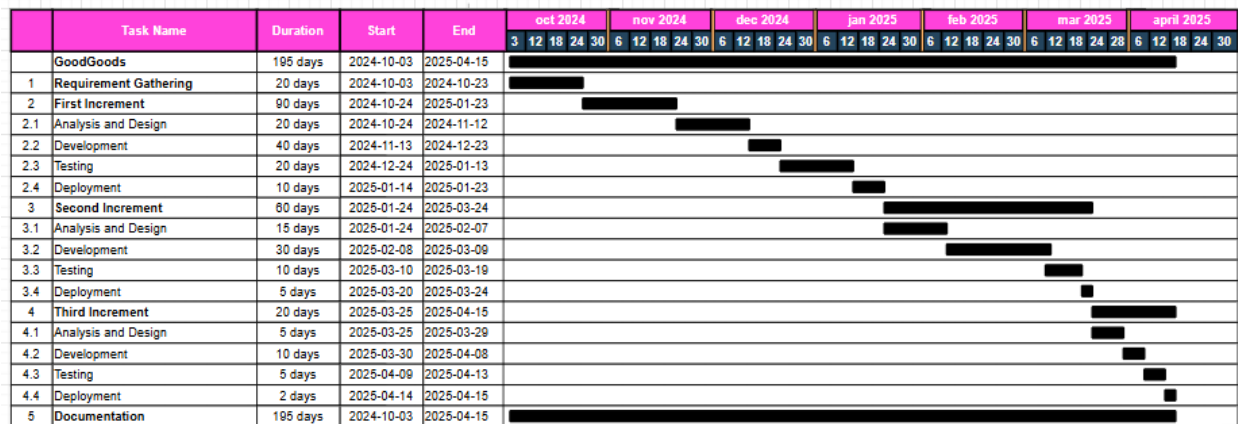


Figure 22 Revised Gantt Chart

5. Further Work

No	Task	Expected End Date	Comment
1	Create Admin and Product Management Pages	30th Jan 2025	Develop an intuitive admin dashboard to manage product listings, track orders, and oversee customer details.
1.1	Build React components for managing products and orders	30th Jan 2025	Design easy-to-use components for adding, editing, and managing product listings and customer orders.
1.2	Create pages for adding and updating products	30th Jan 2025	Design straightforward pages to easily add new products, edit existing ones, and remove outdated listings.
2	Implement Product and Order Management (CRUD)	14th Feb 2025	Set up Create, Read, Update, delete (CRUD) functionalities for managing products and processing orders.
2.1	Set up CRUD operations for products	14th Feb 2025	Develop backend and frontend features to create, update, and delete product listings efficiently.
2.2	Add validation and error handling to	14th Feb 2025	Ensure that product data is validated correctly and handle errors gracefully on both the front and back ends.

	CRUD operations		
3	Develop Search and Filter Features	7th Mar 2025	Create a simple and efficient search bar with filtering options for customers to easily browse products.
3.1	Build Search functionality with product filters	7th Mar 2025	Implement search options that allow customers to filter products by category, price, and other attributes.
3.2	Ensure products show availability status	7th Mar 2025	Display real-time product availability (in stock or out of stock) on the search results page.
4	Develop Shopping Cart and Checkout Process	21st Mar 2025	Build a seamless cart and checkout experience, including a secure and easy payment process.
4.1	Build Cart functionality	21st Mar 2025	Create a shopping cart that allows users to add/remove items, update quantities, and review selections easily.

4.2	Implement Secure Checkout and Payment Gateway	21st Mar 2025	Design a smooth checkout page where customers can review their orders and complete payments securely through a payment gateway.
5	Integrate Payment Gateway	10th Apr 2025	Set up a trusted payment gateway to handle secure transactions on the platform.
5.1	Complete Payment Gateway Integration	10th Apr 2025	Follow integration guidelines for the chosen payment provider to securely process payments.
6	Conduct Testing and Quality Assurance	18th Apr 2025	Test all the critical features such as product listings, cart, checkout, and payment systems to ensure they work flawlessly.
6.1	Perform unit, integration, and load testing	18th Apr 2025	Thoroughly test individual features and the entire system for performance under load and reliability during transactions.

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